SMART INDIA HACKATHON 2025

- **Problem Statement ID** – SIH25104
- **Problem Statement Title -** Developing an AI-powered multilingual chatbot system for seamless student query resolution across college campuses
- **Theme -** Miscellaneous
- **PS Category -** Software
- **Team ID -** 85116
- **Team Name -** Team 100x

Team
100X

LANGUAGE AGNOSTIC CHATBOT

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Proposed Solution

Multilingual Conversations :

- Real-time voice/text in Hindi, English, and **regional languages with automatic language detection**—no setup needed.
- Supports seamless **mid-chat switching** and full coverage of regional dialects, **including Marwari.**

Update Knowledge base :

- Automatically updates responses when colleges publish new policies or circulars.
- Reads PDFs, circulars, FAQs using RAG-based retrieval for **instant, accurate answers.**
- Supports any document type—**PDF, PNG, text**—just like uploading to Google Drive.
- Maintains context across **multi-turn chats for continuous, complex interactions.**

Reminders / Appointment booking :

- Sends **SMS/WhatsApp** alerts for deadlines, updates, and announcements.
- Students **book meetings** with staff using natural language—**auto-updated on calendars.**
- Real-time chatbot needs only name and number—no forms or group subscriptions.
- Complex queries are **handed off to human staff** with full conversation context.

Integration with legacy websites :

- **Plug-and-play integration** with any college website (static or React) in **under 30 seconds**.
- Chatbot goes live with **under five lines of code**, preserving branding via customizable themes.
- Includes interaction logging and admin insights on **student queries, peak usage, and content gaps.**

Link

Feature wise implementation | Abstract Report | Multilingual demo

How it addresses the problem?

- Multilingual chatbot (**Hindi, English + 5 regional languages**) removes queues and language barriers.
- Deploy in **30 seconds with 5 lines** of branded code.
- Students book via chat; staff calendars auto-update with full context.
- Upload policy PDFs—RAG-powered bot answers **instantly and accurately.**

Innovation & Uniqueness

- **No other campus chatbot** offers instant switching between Hindi, English, and regional dialects—no setup needed.
- **Competitors take weeks;** we go live instantly with just 5 lines of code.
- Students say "I **need to meet my counselor tomorrow**"—it's **auto-scheduled**, no forms or apps.
- Our RAG engine **reads PDFs in real-time**, updating itself without human input.

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TECHNICAL APPROACH

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Technologies

AI
AI SDK
Gemini

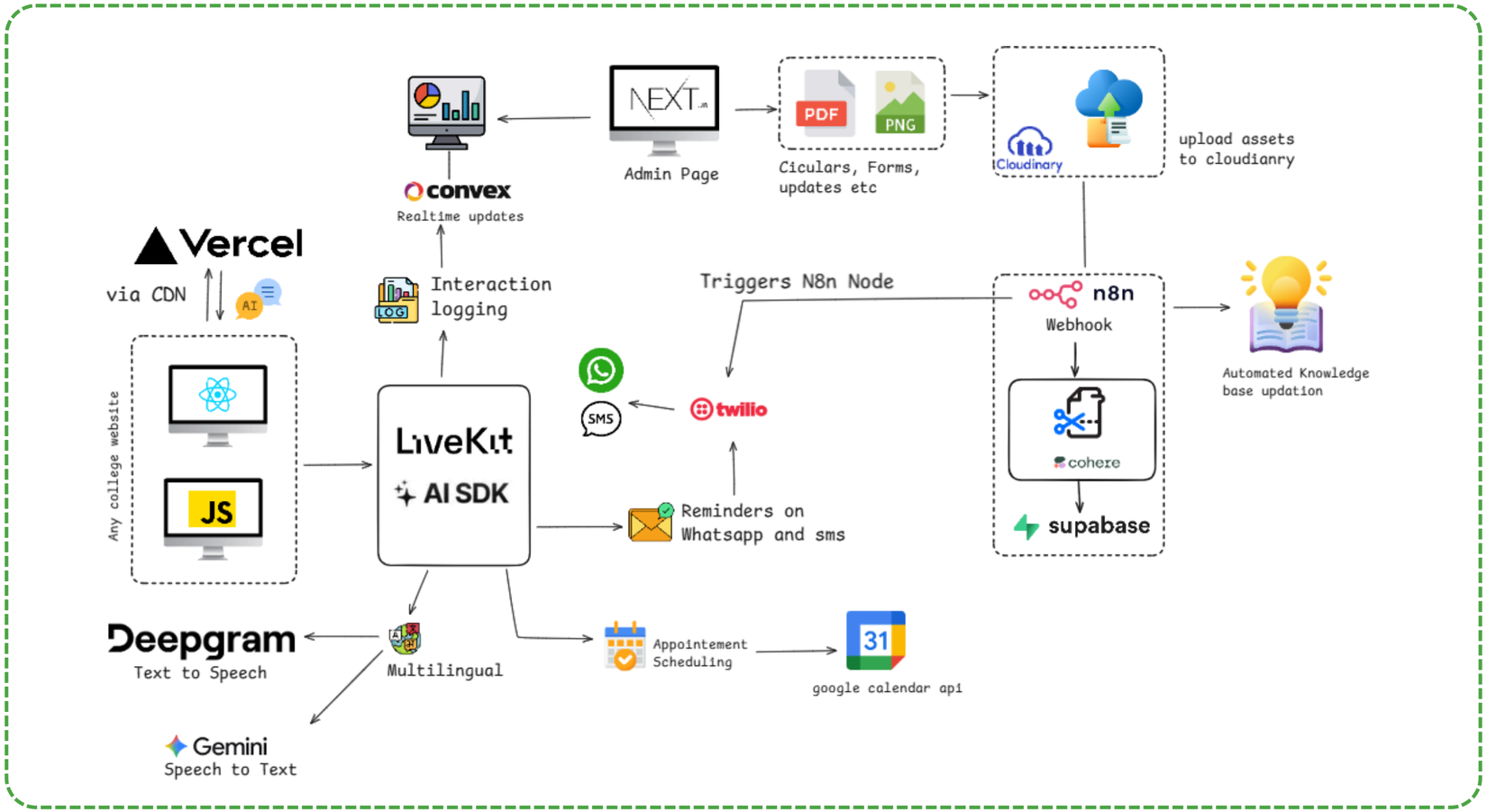
RAG
TypeScript
Pgvector
NEXT.JS
cohere

Multilingual Voice Agent
LiveKit
VAPI
Deepgram
Gemini TTS

Appointment booking and alert
31
twilio
n8n

System Architecture

Link for System Architecture



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FEASIBILITY AND VIABILITY

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Feasibility	Challenges	Mitigation
₹15-25 annual cost compared to traditional helpdesk expenses	Initial setup complexity for non-technical staff	Student volunteer training and detailed documentation
Automated document processing ensures updated knowledge base	Large PDF volumes may degrade processing performance	Chunked processing with optimized RAG architecture
GDPR compliance with built-in encryption protocols	Data privacy concerns for sensitive student information	End-to-end encryption with regular security audits
60% workload reduction freeing staff for complex tasks	Inaccurate responses affecting institutional credibility	Human fallback system with conversation logging

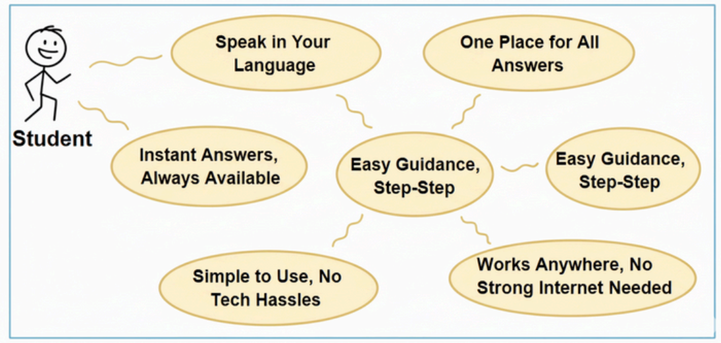
Estimated Technical Costs (per college)

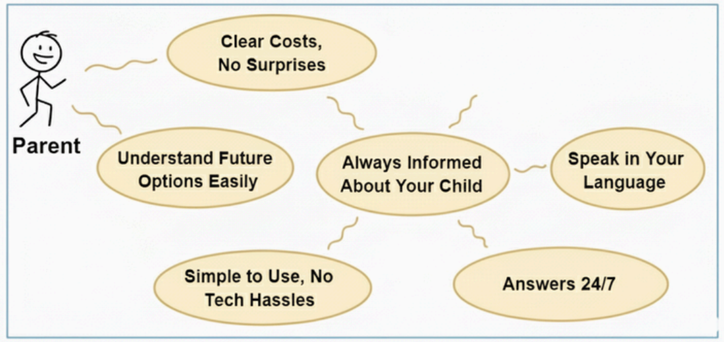
- Cloud serverless function invocation: **₹7–12 per student** annually, with per-chat cost close to **₹0.01–₹0.03 for low/medium usage.**
- **CDN and storage** (for PDFs/circulars): **₹2–3 per student** annually.
- Total annual technology cost per college: ₹15–25 per student (lower if bulk/state contracts negotiated).

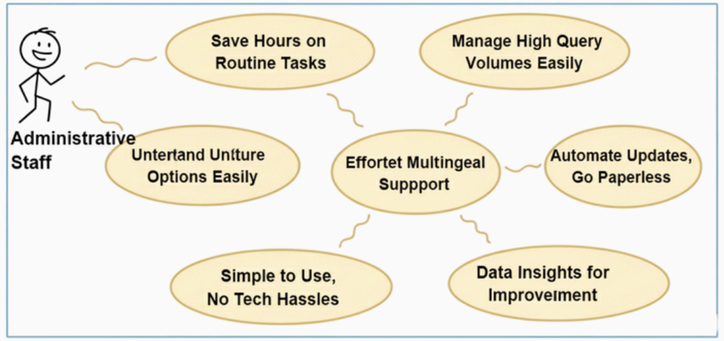
Business Model for State Rollout

- Government either centrally hosts and manages the platform, billing each college at actual usage, or allows colleges to subscribe with **tiered plans** (e.g., **₹10,000 starting fee + ₹15/student/year**).
- **Support and admin dashboard** offered as a value-add, allowing **DTE staff to monitor usage, log stats, and update responses from a central console.**
- The **state educational budget** can easily absorb these costs compared to hiring additional helpdesk staff or running large-scale call centers.

Use Case Diagram







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IMPACT AND BENEFITS

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IMPACT BENEFITS

1. **Massive Cost Advantage**
Our AI-powered chatbot slashes **costs by 75%**, charging just **₹15–25 per student** annually versus ₹100–150 on enterprise platforms. Institutions **save ₹8–12 lakhs per year**, freeing funds for student growth.

2. **Proven Administrative Efficiency**
Automation reduces administrative overhead by **30%, eliminating** overtime during peak admissions and cutting the need for additional staff infrastructure.

3. **Drastic Query Resolution Boost**
Institutions shift from 500+ manual staff hours daily to automated **24/7 service**—delivering **67% higher efficiency** and maintaining 92% accuracy.

4. **Lightning-Fast Response Times**
Queries that used to **take 2–3 days** now get resolved in **under 2 seconds**, satisfying the **63% of students** who want round-the-clock support.

1 3X Student Satisfaction

24/7 multilingual support (Hindi, English + regional dialects) creates happier students, directly boosting retention and reducing dropout.

2 Application Success Lift

Guided chatbot support improves form completion rates by **40%**, ensuring fewer dropouts during admissions and scholarship processes.

3 Rural Access Unlocked

Voice-first design taps into an untapped **40% rural market,** reaching students left behind by English-only, text-heavy platforms.

4 Plug & Play Simplicity

Integrates with any college website in under **30 seconds** using fewer than 5 lines of code—no complex setup, no rebranding issues.

5 Always Accurate, Always Updated

RAG-based knowledge system automatically syncs with new circulars and policies, ensuring **92% accurate, up-to-date** answers without manual effort.

6 Policy-Aligned & Secure

Supports NEP 2020 and Digital India goals, with **DPDP Act compliance** for data privacy—building trust with institutions and government.

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RESEARCH AND REFERENCES

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1. Systematic Approach to Evaluate Chatbots in Educational Contexts (JISEM, 2025)

   Gamification and Chatbots in Education: Impact on Student Engagement

2. Digital Transformation for Educational Chatbots: Design & Implementation (IEEE, 2022)

   Public chatbot implementation methodology and architecture design

3. How to Improve Government Efficiency Through Digital Solutions (Wavetec, 2024)

   Digital transformation trends and queue management automation for government services

4. Custom-Trained Large Language Models as Educational Resources (MDPI Croatia, 2024)

   Exploring custom LLM applications in business management education with multilingual support

5. EduBot (Andhra University)

   English-only, text-based interface, single university focus

6. Multilingual College Admission Bot (Jain University)

   Text-only, requires constant internet, college-specific implementation